Cross-Scene Diffusion-Enhanced Uncertainty Attention Network for Hyperspectral Image Classification

Chenglong Wang, Yi Guo, Minchao Ye, *Member, IEEE*, and Fengchao Xiong, *Member, IEEE*

***Abstract*—Cross-scene hyperspectral image (HSI) classification is widely used for its exceptional ability to address the small-sample-size problem. However, important challenges remain, including domain shift and noise between scenes, the need to align features while accounting for sample-level reliability, and the difficulty of fusing domain-invariant and scene-specific features without degrading discriminability. To enhance generalization and reliability with small sample sizes, we propose a cross-scene diffusion-enhanced uncertainty attention network (CDUAN). CDUAN consists of three main components: 1) a domain adaptation diffusion module (DADM) that utilizes target-guided diffusion to synthesize augmented source samples, establishing distribution-level alignment as the first step; 2) a cross-scene uncertainty-aware mutual attention module (CUMAM) is then proposed to ensure feature-level reliability control during alignment across scenes by integrating a sample-level uncertainty mechanism. This mechanism adjusts attention weights to highlight domain-invariant features and, during fusion, retain as much discriminative information as possible; and 3) a cross-scene fusion attention module (CSFAM) realizes fusion-level control of invariance and discriminability across scenes by combining multiscale contrastive pairing with attention-based fusion. CSFAM maximizes relevant cross-scene feature similarity, suppresses noise, and ensures discriminative representation for accurate classification. Extensive experiments on three cross-scene datasets validate the effectiveness of the proposed CDUAN.**

***Index Terms*—Cross-scene attention mechanism, cross-scene hyperspectral image (HIS) classification, data augmentation, domain adaptation diffusion module (DADM).**

Received 23 October 2025; revised 13 January 2026 and 11 February 2026; accepted 22 March 2026. Date of publication 25 March 2026; date of current version 10 April 2026. This work was supported by the Science and Technology Program Projects of Shanghai Municipal Committee of Science and Technology under Grant 22DZ204903 and Grant 22511104800. *(Corresponding author: Yi Guo.)*

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Digital Object Identifier 10.1109/TGRS.2026.3677462

I. Introduction

HYPERSPECTRAL image (HSI), with its rich spectral dimensional information, enables precise identification of ground truths, thus finding extensive applications in remote sensing [1], [2], agriculture [3], and environmental monitoring [4]. Despite the progress in single-scene classification, limited labeled samples remain a major obstacle due to the high cost and complexity of data collection in real-world scenes. Consequently, it is common practice to leverage a scene with abundant labeled samples (source scene) to assist a scene with limited labeled samples (target scene), a method known as cross-scene HSI classification [5], [6]. However, cross-scene HSI classification poses numerous challenges, primarily due to spectral feature discrepancies between different scenes. These discrepancies are mainly attributed to varying environmental conditions [7], sensor calibration [8], and changes in spatial resolution [9]. For instance, various weather conditions, seasonal variations, and atmospheric states can significantly impact the spectral response of HSI. Moreover, sensor calibration inconsistencies over time and under different environmental conditions, as well as sensor aging, can lead to variations in the acquired spectral data.

Domain adaptation (DA) addresses the challenge of differing distributions between the source and target scenes. It aims to bridge the gap between these scenes so that training data from the source scene can be effectively leveraged to enhance performance in the target scene. In recent years, DA has become increasingly important and is now widely applied in cross-scene HSI classification [10], [11], [12]. Zhang et al. [13] proposed a network architecture that integrates topological structure information with semantic information transfer to achieve DA across different scenes. In particular, this approach involves constructing a graph neural network to extract the topological information of each scene and employing a semantic transfer mechanism to align the feature representations across different scenes, thereby enhancing the model's generalization ability in cross-scene classification tasks. Ye et al. [14] developed a cross-scene mapping chain to facilitate DA, introducing a method that employs a multi-CycleGAN framework. This method focuses on learning to map images between the source and target domains in both directions, using cycle consistency to ensure effective domain transformation and feature alignment. Yang et al. [15] leveraged graph embedding techniques to capture the structural information between

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the source and target domains, and employed an interclass relation-aware mechanism to align the feature representations across domains. This approach effectively reduces domain discrepancies, thereby achieving DA.

Although DA methods have shown strong performance in cross-scene HSI classification, they have certain limitations. Many of these methods focus on aligning global features, often at the expense of overlooking subtle, fine-grained differences. As a result, they may fail to capture and emphasize critical features when dealing with complex datasets. In addition, these methods often struggle to detect and adapt to variations in local features, leading to reduced accuracy and robustness in more challenging scenarios. To address these issues, the cross-scene attention mechanism has been proposed in recent years. This mechanism not only improves the alignment of global features but also enhances the focus on fine-grained differences and effectively captures local feature variations. Mdrafi et al. [16] utilized a residual network to capture deep features within HSIs and employed a cross-scene attention mechanism to dynamically align features between the source and target domains. This cross-scene attention mechanism focuses on features that significantly contribute to cross-scene classification while diminishing the impact of irrelevant or noisy features. By accurately detecting subtle variations between scenes, the approach effectively reduces differences between different scenes. Zhu et al. [17] proposed a cross-scene spatial attention mechanism that dynamically adjusts the weights of different regions, combined with deformable convolution to enhance the model's flexibility and adaptability. This approach not only effectively captures the relevant features between the source and target scenes but also improves the model's performance in handling geometric transformations and shape variations across different scenes. Cai et al. [18] proposed a multilevel unsupervised DA (MLUDA) network that incorporates alignment at the image, feature, and logic levels. During the feature alignment process, they introduced a multibranch cross-attention (MBCA) mechanism to dynamically adjust feature weights, allowing for better alignment of the most relevant features.

Currently, although cross-scene attention mechanisms perform well in feature alignment, they are often prone to issues like local optima and sensitivity to noise. In particular, attention mechanisms may overemphasize locally significant features while neglecting global information, leading to insufficient generalization in complex or variable target domains. In addition, noise or irrelevant features can interfere with the attention mechanism's judgment, resulting in unstable alignment. To mitigate these challenges, diffusion networks are regarded as a promising solution. By enabling global feature propagation, diffusion models can circumvent local optima and attenuate the influence of noise, thereby enhancing the robustness and overall performance of the model. In recent years, diffusion networks have seen widespread application in the field of computer vision [19], [20], [21], [22]. Cao et al. [23] leveraged a diffusion model to propose an image enhancement algorithm to improve the clarity of HSIs. In particular, the diffusion process simulates multiscale feature propagation, effectively preserving and enhancing detailed information within the images, thereby avoiding local optima. In addition, the study introduces a disentangled modulation mechanism to separate different feature layers within the images, enabling the diffusion model to more accurately process information across various spectral bands and reduce the impact of noise. Huang et al. [24] proposed a wavelet-based diffusion model (WaveDM) for image restoration. This method combines the strengths of wavelet transforms and diffusion models, effectively reducing noise and progressively restoring degraded or damaged images through multiscale image processing. Zhu et al. [25] proposed a high- and low-frequency progressive-guided diffusion model for classifying panchromatic (PAN) and multispectral (MS) images. This model employs a gradual, multistep process to ensure the effective integration of high- and low-frequency information, thereby significantly enhancing image classification performance.

In summary, existing attention-based DA methods suffer from several limitations. Under scarce labels, domain shift remains large because common augmentations and shallow statistics fail to move the source distribution toward the target. Attention often treats features as equally reliable, allowing noisy or low-confidence features to influence alignment and push optimization into local optima. During fusion, discriminative features are frequently averaged away in pursuit of invariance, which weakens class separability. These limitations lead to unstable alignment and poor generalization in complex target scenes. To address these issues, we propose a novel DA framework called the cross-scene diffusion-enhanced uncertainty attention network (CDUAN). CDUAN integrates the strengths of a cross-scene diffusion model with a cross-scene attention mechanism to achieve effective feature alignment, which refers to aligning feature distributions between the source and target scenes to reduce domain discrepancies. This combination not only ensures feature invariance across different scenes but also enhances the model's ability to generalize effectively to new environments. The main contributions of CDUAN are as follows.

1) DA diffusion module (DADM) improves the reliability of source training samples by modeling a diffusion process that gradually transfers source features toward the target scene distribution, ensuring better cross-scene learning. This module facilitates source samples becoming more similar to target samples.
2) Cross-scene uncertainty-aware mutual attention module (CUMAM) computes a specially designed variance for each sample feature and uses it to weight that feature's contribution, achieving more than naive alignment by suppressing noisy and shifted samples and strengthening cross-scene consistency.
3) Cross-scene fusion attention module (CSFAM) maintains feature invariance across scenes by combining multiscale contrastive learning with cross-scene attention mechanisms, where the multiscale contrastive component captures both global and local semantic relationships by contrasting features at different spatial scales, thereby further enhancing the consistency and transferability of the learned representations across scenes.

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This article is structured as follows. Section II provides an overview of diffusion models and attention mechanisms. Section III details the design and implementation of the proposed CDUAN. Section IV presents our experimental setup and offers a thorough analysis of the results. Finally, Section V summarizes the key findings and conclusions of the study.

II. Related Work

The proposed CDUAN effectively mitigates the impact of scene differences by integrating a cross-scene diffusion model and a cross-scene attention mechanism. This approach enhances the model's robustness and performance in complex environments. In this section, we will provide a detailed introduction to the diffusion model and the attention mechanism.

A. Diffusion Model

Diffusion models are a significant class of probabilistic generative models, first introduced by Sohl-Dickstein et al. [26] and have quickly become a core technology for data generation and transformation tasks. The fundamental principle of diffusion models is to transform complex data distributions into simple noise distributions through a series of random steps, followed by a reverse learning process to restore the original data distribution. This iterative method allows diffusion models to accurately capture complex patterns and structures in data, making them particularly effective in generating high-quality samples. Initially, diffusion models were widely applied in image [27] and video generation [28], aiming to create realistic visual samples from learned distributions.

In recent years, diffusion models have become widely utilized across numerous fields due to their exceptional ability to generate high-quality samples. In natural language processing (NLP), diffusion models have been employed for text generation [29], language translation [30], and dialog systems [31], demonstrating remarkable success in generating coherent and contextually relevant text. In addition, in the field of audio processing, diffusion models have shown significant potential in tasks such as speech synthesis [32], audio restoration [33], and music generation [34], producing highly realistic and detailed audio outputs. Furthermore, diffusion models have been extensively applied to HSI classification [35], [36], effectively handling the complex spectral information inherent in such data.

However, despite their strong performance, traditional diffusion models are typically trained on specific scenarios, limiting their capacity to generalize beyond the training data distribution. In particular, these models struggle to generate data that lies outside their training distribution, leading to decreased reliability in their outputs when faced with novel or unseen scenarios. To address these limitations and ensure the generation of high-quality data across varying domains, cross-scene diffusion models have gained increasing attention and usage. Wang and Wang [37] developed a frequency-domain diffusion-based generative module to effectively reduce noise and enhance speech quality by isolating and minimizing interference. Peng et al. [38] introduced a domain-adaptive mechanism into the diffusion model, enabling the unlabeled target domain to be effectively aligned with the source domain. Yang et al. [39] proposed a cross-scene text-to-image diffusion model that effectively mitigates domain differences by aligning the semantic features of text with the visual features of images, enhancing the quality and accuracy of generated images.

In cross-scene HSI classification, having access to a large volume of high-quality data is crucial for effective model adaptation. Inspired by the aforementioned methods, we employ a cross-scene diffusion model to address the challenges associated with limited training samples in small-sample-size scenarios.

B. Attention Mechanism

The attention mechanism has gained significant attention in image classification in recent years. The core of the attention mechanism is to mimic the human attention process when dealing with complex and massive amounts of information. In image classification, it dynamically allocates computational resources and focus points based on the relative importance of different input data or features. Initially, CNNs were employed to enhance the recognition capabilities for important areas of an image. With the development of the Transformer model, self-attention mechanisms have been utilized extensively. For instance, Ding et al. [40] adopted a weakly supervised pyramid attention network that consists of multiple convolution and attention layers. Each attention layer extracts information from the previous layer, effectively addressing the challenges of fine-grained image classification [40]. Ji et al. [41] extended the self-attention mechanism to a polarized self-attention approach to emphasize the importance of specific spatial–spectral features for HSI classification.

However, in reality, training an efficient model often requires a large amount of data and computational resources, making it crucial to enhance the model's generalization capability. Traditional single-source attention mechanisms, which rely on a single data source, may suffer from insufficient generalization because features across different domains can vary significantly. The cross-scene attention mechanism has been increasingly mentioned recently to address this issue. For example, Huang et al. [42] introduced a cross-scene attention mechanism that computes the similarity between source and target scene features to dynamically adjust the attention weights of each domain's features, achieving effective feature alignment. Wang et al. [43] proposed a specially designed cross-scene loss to increase the separation between interclass samples while preserving the similarity within intraclass samples, and so on.

Inspired by the cross-scene attention mechanism, we developed the CDUAN, which integrates a cross-scene diffusion model with a cross-scene attention mechanism. In Section III, we will show the details of the proposed CDUAN.

III. Proposed CDUAN

Typically, the training samples from the source scene $\mathcal{S}_i \in \mathbb{N}^{w\times w\times ch_{\mathcal{S}}}, i \in \{1,2,\ldots,m\}$ differ in size from those of the target scene $\mathcal{T}_j \in \mathbb{N}^{w\times w\times ch_{\mathcal{T}}}, j \in \{1,2,\ldots,n\}$. Here, $\mathcal{S}$ and $\mathcal{T}$ denote the source and target scenes, with m and n representing

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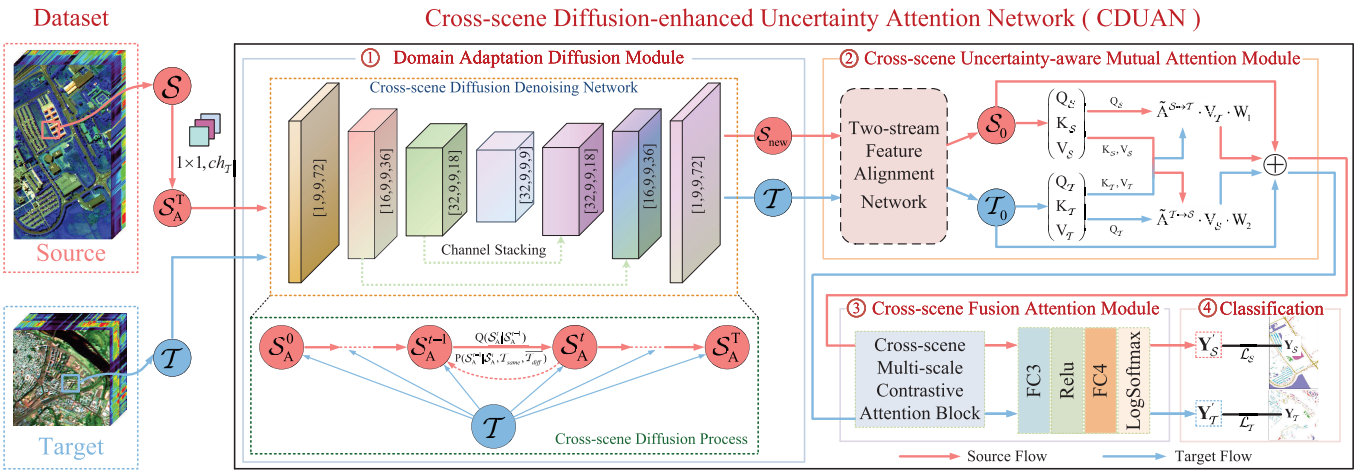


Fig. 1. Architecture of the whole CDUAN.

the number of samples from each scene, respectively, while $w \times w$ represents the height and width of the selected patch. The architecture of our proposed CDUAN is shown in Fig. 1. The CDUAN effectively addresses the challenge of varying feature channel sizes across scenes by applying a $1 \times 1, ch_{\mathcal{T}}$ convolution to the source scene training samples. This operation ensures the channel sizes are aligned between the source and target scenes. After aligning the input size, the DADM generates high-quality synthetic data for data augmentation under the constraint of target scene samples, thereby enhancing the network's generalization capabilities. Furthermore, an uncertainty-aware mutual attention module is employed to extract aligned spatial–spectral features according to a cross-scene attention mechanism. With a meticulously designed cross-scene fusion attention block, the shift between two scenes is significantly reduced.

A. DA Diffusion Module

In cross-scene classification tasks, it is essential to maximize the utility of information extracted from the source scene training samples. Ideally, these samples should closely mirror those of the target scene while retaining distinctive features that allow the model to generalize effectively across diverse scenes. Achieving this balance requires leveraging the target scene training samples to refine and enhance the relevance of the source data. To address this challenge, we developed the DADM, a method designed to generate more effective source training samples for subsequent processing. The DADM integrates the original target training samples $\mathcal{T}$ with source samples $\mathcal{S}_{\mathrm{A}}^{\mathrm{T}}$, which have been augmented with Gaussian noise. The cross-scene diffusion model then synthesizes new source training samples that are both high-quality and enriched with critical features. Before introducing the cross-scene diffusion model, we begin by exploring the foundational diffusion process.

Here, we illustrate the diffusion model using a training sample s. The diffusion model generates data through a bidirectional process: the forward diffusion process $\mathrm{Q}(s^t|s^{t-1})$ progressively adds Gaussian noise to the data, transforming it into pure noise, while the reverse diffusion process $\mathrm{P}(s^{t-1}|s^t)$ uses a neural network to gradually denoise and reconstruct the original data. The noisy sample at timestep t, s^t, is obtained from s^{t-1} through the Markov transition

$$\mathrm{Q}\left(s^t \middle| s^{t-1}\right) = \mathcal{N}\left(s^t;\ \sqrt{1-\beta_t}s^{t-1}, \beta_t \mathbf{I}\right) \tag{1}$$

where $\mathcal{N}$ denotes the Gaussian distribution, β_t controls the noise level at timestep t and follows a variance schedule $\beta_t \in (0,1)$ for $t = 0, \ldots, T$, with T representing the total number of timesteps. Here, $\mathbf{I}$ refers to the identity matrix, which represents the standard normal distribution. Using (1), the sample s^t at timestep t can be expressed as

$$s^t = \sqrt{\bar{\alpha}_t}s_0 + \sqrt{1-\bar{\alpha}_t}\varepsilon, \varepsilon \sim \mathcal{N}(0,1) \tag{2}$$

where ε represents the Gaussian noise added at each timestep. Here, $\alpha_t = 1 - \beta_t$, and $\bar{\alpha}_t = \Pi_{i=1}^{t}\alpha_i$ denotes the cumulative product of α values from timestep 1 to t. Similar to the forward process, the inverse process, which seeks to recover the noisy sample s^t from Gaussian noise ε to reconstruct s_0, is also a Markov process. However, due to the noise's impact, the transition probability $\mathrm{P}(s^{t-1}|s^t)$ cannot be directly estimated. To address this, a diffusion denoising network is utilized to approximate the conditional probability distribution by predicting the added noise ε, represented as

$$\mathrm{P}\left(s^{t-1}|s^t\right) = \mathcal{N}\left(s^{t-1}; \mu\left(s^t\right), \sigma^2\left(s^t\right)\mathbf{I}\right) \tag{3}$$

where $\mathrm{P}(s^{t-1}|s^t)$ represents the inferred inverse transition probability, with $\mu(s^t)$ and $\sigma^2(s^t)$ as the mean and variance, respectively, of the predicted distribution.

Next, we introduce the cross-scene diffusion model we propose. Traditional diffusion processes are typically based on single-scene training samples, which may not adequately address challenges from varying scenes. Therefore, it is essential to incorporate information from other scenes when generating new samples. In particular, during the reverse diffusion process, target scene samples should also be considered, which can be expressed as $\mathrm{P}(\mathcal{S}_{\mathrm{A}}^{t-1}|\mathcal{S}_{\mathrm{A}}^{t}, \mathcal{T}_{same}, \overline{\mathcal{T}_{diff}})$. The $\mathcal{T}_{same}$ and $\overline{\mathcal{T}_{diff}}$ refer to target samples that belong to the same class and different classes, respectively, when compared to the currently generated source sample. The index for the same class is denoted as k, while the index for different classes is

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denoted as v. This distinction is crucial for effectively guiding the generation process, ensuring that the source samples either align closely with target samples of the same class or distinctly differ from those of different classes, depending on the objective of the generation process. This ensures that the model adapts better across different scenes. Following this, we first provide a detailed derivation of the cross-scene diffusion model with only the same class target samples $\mathrm{P}(\mathcal{S}_{\mathrm{A}}^{t-1}|\mathcal{S}_{\mathrm{A}}^{t}, \mathcal{T}_{same})$. Using Bayes' theorem, the expression can be derived as follows:

$$\mathrm{P}\left(\mathcal{S}_{\mathrm{A}}^{t-1}\middle|\mathcal{S}_{\mathrm{A}}^{t}, \mathcal{T}_{same}\right) = \frac{\mathrm{P}\left(\mathcal{S}_{\mathrm{A}}^{t-1}\middle|\mathcal{S}_{\mathrm{A}}^{t}\right)\mathrm{P}\left(\mathcal{T}_{same}\middle|\mathcal{S}_{\mathrm{A}}^{t-1}, \mathcal{S}_{\mathrm{A}}^{t}\right)}{\mathrm{P}\left(\mathcal{T}_{same}\middle|\mathcal{S}_{\mathrm{A}}^{t}\right)} \tag{4}$$

where each $\mathcal{T}_i$ is conditionally independent given $\mathcal{S}_{\mathrm{A}}^{t-1}$ and $\mathcal{S}_{\mathrm{A}}^{t}$, so the (4) can be rewritten as

$$\mathrm{P}\left(\mathcal{S}_{\mathrm{A}}^{t-1}\middle|\mathcal{S}_{\mathrm{A}}^{t}, \mathcal{T}_{same}\right) = \frac{\mathrm{P}\left(\mathcal{S}_{\mathrm{A}}^{t-1}\middle|\mathcal{S}_{\mathrm{A}}^{t}\right)\prod_{i\in k}^{len(k)}\mathrm{P}\left(\mathcal{T}_i\middle|\mathcal{S}_{\mathrm{A}}^{t-1}, \mathcal{S}_{\mathrm{A}}^{t}\right)}{\prod_{i\in k}^{len(k)}\mathrm{P}\left(\mathcal{T}_i\middle|\mathcal{S}_{\mathrm{A}}^{t}\right)}. \tag{5}$$

Then, we suppose the $\mathrm{P}(\mathcal{T}_i|\mathcal{S}_{\mathrm{A}}^{t-1}, \mathcal{S}_{\mathrm{A}}^{t}) = \mathrm{P}(\mathcal{T}_i|\mathcal{S}_{\mathrm{A}}^{t-1})$, i.e., the current source scene $\mathcal{S}_{\mathrm{A}}^{t}$ is generated by applying task-irrelevant noise or transformation to $\mathcal{S}_{\mathrm{A}}^{t-1}$. Since such perturbations do not carry additional semantic information relevant to the target sample $\mathcal{T}_i$, the influence of $\mathcal{S}_{\mathrm{A}}^{t}$ becomes negligible. Under the assumption of a smooth and continuous domain shift, this simplification facilitates a tractable model design while preserving the essential dependence structure. So the equation can be simplified as follows:

$$\mathrm{P}\left(\mathcal{S}_{\mathrm{A}}^{t-1}\middle|\mathcal{S}_{\mathrm{A}}^{t}, \mathcal{T}_{same}\right) = \frac{\mathrm{P}\left(\mathcal{S}_{\mathrm{A}}^{t-1}\middle|\mathcal{S}_{\mathrm{A}}^{t}\right)\prod_{i\in k}^{len(k)}\mathrm{P}\left(\mathcal{T}_i\middle|\mathcal{S}_{\mathrm{A}}^{t-1}\right)}{\prod_{i\in k}^{len(k)}\mathrm{P}\left(\mathcal{T}_i\middle|\mathcal{S}_{\mathrm{A}}^{t}\right)}. \tag{6}$$

In this equation, $\mathrm{P}(\mathcal{T}_i|\mathcal{S}_{\mathrm{A}}^{t})$ does not rely on $\mathcal{S}_{\mathrm{A}}^{t-1}$, thus the denominator can be regarded as a constant. Therefore, the posterior is proportional to

$$\begin{aligned}\mathrm{P}\left(\mathcal{S}_{\mathrm{A}}^{t-1}\middle|\mathcal{S}_{\mathrm{A}}^{t}, \mathcal{T}_{same}\right) &= \mathrm{P}\left(\mathcal{S}_{\mathrm{A}}^{t-1}\middle|\mathcal{S}_{\mathrm{A}}^{t}\right)\\ &\cdot \exp\left(\sum_{i\in k}\log\mathrm{P}\left(\mathcal{T}_i\middle|\mathcal{S}_{\mathrm{A}}^{t-1}\right) - \sum_{i\in k}\log\mathrm{P}\left(\mathcal{T}_i\middle|\mathcal{S}_{\mathrm{A}}^{t}\right)\right).\end{aligned} \tag{7}$$

Then, we perform a Taylor expansion for each $\log\mathrm{P}(\mathcal{T}_i|\mathcal{S}_{\mathrm{A}}^{t-1})$. Considering that $\mathcal{S}_{\mathrm{A}}^{t-1}$ and $\mathcal{S}_{\mathrm{A}}^{t}$ are very close (due to the small timestep), we expand to the first-order term

$$\begin{aligned}\log\mathrm{P}\left(\mathcal{T}_i|\mathcal{S}_{\mathrm{A}}^{t-1}\right) &\approx \log\mathrm{P}\left(\mathcal{T}_i|\mathcal{S}_{\mathrm{A}}^{t}\right)\\ &+ \left(\mathcal{S}_{\mathrm{A}}^{t-1} - \mathcal{S}_{\mathrm{A}}^{t}\right)\nabla_{\mathcal{S}_{\mathrm{A}}^{t}}\log\mathrm{P}\left(\mathcal{T}_i|\mathcal{S}_{\mathrm{A}}^{t}\right).\end{aligned} \tag{8}$$

Substituting this into (7), and absorbing the terms independent of $\mathcal{S}_{\mathrm{A}}^{t-1}$ into the proportionality constant, we obtain

$$\begin{aligned}\mathrm{P}\left(\mathcal{S}_{\mathrm{A}}^{t-1}\middle|\mathcal{S}_{\mathrm{A}}^{t}, \mathcal{T}_{same}\right) &= \mathrm{P}\left(\mathcal{S}_{\mathrm{A}}^{t-1}\middle|\mathcal{S}_{\mathrm{A}}^{t}\right)\\ &\cdot \exp\left(\sum_{i\in k}\left[\log\mathrm{P}\left(\mathcal{T}_i|\mathcal{S}_{\mathrm{A}}^{t}\right) + \left(\mathcal{S}_{\mathrm{A}}^{t-1} - \mathcal{S}_{\mathrm{A}}^{t}\right)\nabla_{\mathcal{S}_{\mathrm{A}}^{t}}\right.\right.\\ &\left.\left.\log\mathrm{P}\left(\mathcal{T}_i|\mathcal{S}_{\mathrm{A}}^{t}\right)\right] - \sum_{i\in k}\log\mathrm{P}\left(\mathcal{T}_i|\mathcal{S}_{\mathrm{A}}^{t}\right)\right)\\ &= \mathrm{P}\left(\mathcal{S}_{\mathrm{A}}^{t-1}\middle|\mathcal{S}_{\mathrm{A}}^{t}\right)\cdot\exp\left(\left(\mathcal{S}_{\mathrm{A}}^{t-1} - \mathcal{S}_{\mathrm{A}}^{t}\right)\sum_{i\in k}\nabla_{\mathcal{S}_{\mathrm{A}}^{t}}\log\mathrm{P}\left(\mathcal{T}_i|\mathcal{S}_{\mathrm{A}}^{t}\right)\right).\end{aligned} \tag{9}$$

According to (3), $\mathrm{P}(\mathcal{S}_{\mathrm{A}}^{t-1}|\mathcal{S}_{\mathrm{A}}^{t})$ is modeled as a Gaussian

$$\begin{aligned}\mathrm{P}\left(\mathcal{S}_{\mathrm{A}}^{t-1}\middle|\mathcal{S}_{\mathrm{A}}^{t}\right) &= \mathcal{N}\left(\mathcal{S}_{\mathrm{A}}^{t-1}; \mu\left(\mathcal{S}_{\mathrm{A}}^{t}\right), \sigma^2\left(\mathcal{S}_{\mathrm{A}}^{t}\right)\mathbf{I}\right)\\ &= \exp\left(-\frac{\left|\left|\mathcal{S}_{\mathrm{A}}^{t-1} - \mu\left(\mathcal{S}_{\mathrm{A}}^{t}\right)\right|\right|^2}{2\sigma^2\left(\mathcal{S}_{\mathrm{A}}^{t}\right)}\right).\end{aligned} \tag{10}$$

Combining this with (9), we derive the final expression

$$\begin{aligned}&\mathrm{P}\left(\mathcal{S}_{\mathrm{A}}^{t-1}\middle|\mathcal{S}_{\mathrm{A}}^{t}, \mathcal{T}_{same}\right)\\ &= \exp\left(-\frac{\left|\left|\mathcal{S}_{\mathrm{A}}^{t-1} - \mu\left(\mathcal{S}_{\mathrm{A}}^{t}\right)\right|\right|^2}{2\sigma^2\left(\mathcal{S}_{\mathrm{A}}^{t}\right)}\right.\\ &\qquad\left. + \left(\mathcal{S}_{\mathrm{A}}^{t-1} - \mathcal{S}_{\mathrm{A}}^{t}\right)\sum_{i\in k}\nabla_{\mathcal{S}_{\mathrm{A}}^{t}}\log\mathrm{P}\left(\mathcal{T}_i|\mathcal{S}_{\mathrm{A}}^{t}\right)\right)\\ &= \exp\left(-\frac{\left|\left|\mathcal{S}_{\mathrm{A}}^{t-1} - \mu\left(\mathcal{S}_{\mathrm{A}}^{t}\right)\right|\right|^2}{2\sigma^2\left(\mathcal{S}_{\mathrm{A}}^{t}\right)}\right.\\ &\qquad\left. + \sum_{i\in k}\left[\left(\mathcal{S}_{\mathrm{A}}^{t-1} - \mu\left(\mathcal{S}_{\mathrm{A}}^{t}\right)\right)\nabla_{\mathcal{S}_{\mathrm{A}}^{t}}\log\mathrm{P}\left(\mathcal{T}_i|\mathcal{S}_{\mathrm{A}}^{t}\right)\right]\right).\end{aligned} \tag{11}$$

Thus, we have $\mathrm{P}(\mathcal{S}_{\mathrm{A}}^{t-1}|\mathcal{S}_{\mathrm{A}}^{t}, \mathcal{T}_{same})$ is given by

$$\mathcal{N}\left(\mathcal{S}_{\mathrm{A}}^{t-1}; \mu\left(\mathcal{S}_{\mathrm{A}}^{t}\right) + \sigma^2\left(\mathcal{S}_{\mathrm{A}}^{t}\right)\sum_{i\in k}\nabla_{\mathcal{S}_{\mathrm{A}}^{t}}\log\mathrm{P}\left(\mathcal{T}_i|\mathcal{S}_{\mathrm{A}}^{t}\right), \sigma^2\left(\mathcal{S}_{\mathrm{A}}^{t}\right)\mathbf{I}\right). \tag{12}$$

Then, we expand the formulation by incorporating $\overline{\mathcal{T}_{diff}}$, representing different-class samples. Unlike the above derivation, here we aim to ensure that when generating a specific class of source training samples, those from different classes should be as distinct as possible from the generated data. Thus, the expression for $\mathrm{P}(\mathcal{S}_{\mathrm{A}}^{t-1}|\mathcal{S}_{\mathrm{A}}^{t}, \mathcal{T}_{same}, \overline{\mathcal{T}_{diff}})$ follows:

$$\begin{aligned}&\mathrm{P}\left(\mathcal{S}_{\mathrm{A}}^{t-1}\middle|\mathcal{S}_{\mathrm{A}}^{t}, \mathcal{T}_{same}, \overline{\mathcal{T}_{diff}}\right)\\ &= \mathcal{N}\left(\mathcal{S}_{\mathrm{A}}^{t-1}; \mu\left(\mathcal{S}_{\mathrm{A}}^{t}\right) + \sigma^2\left(\mathcal{S}_{\mathrm{A}}^{t}\right)\right.\\ &\qquad\left.\lambda_t\left(\hat{g}_{same}\left(\mathcal{S}_{\mathrm{A}}^{t}\right) - \rho\,\hat{g}_{diff}\left(\mathcal{S}_{\mathrm{A}}^{t}\right)\right), \sigma^2\left(\mathcal{S}_{\mathrm{A}}^{t}\right)\mathbf{I}\right)\end{aligned} \tag{13}$$

where

$$\begin{aligned}g_{same}\left(\mathcal{S}_{\mathrm{A}}^{t}\right) &= \sum_{i\in k}\nabla_{\mathcal{S}_{\mathrm{A}}^{t}}\log\mathrm{P}\left(\mathcal{T}_i|\mathcal{S}_{\mathrm{A}}^{t}\right)\\ g_{diff}\left(\mathcal{S}_{\mathrm{A}}^{t}\right) &= \sum_{j\in v}\nabla_{\mathcal{S}_{\mathrm{A}}^{t}}\log\mathrm{P}\left(\mathcal{T}_j|\mathcal{S}_{\mathrm{A}}^{t}\right)\end{aligned} \tag{14}$$

$$\hat{g}\left(\mathcal{S}_{\mathrm{A}}^{t}\right) = \frac{g\left(\mathcal{S}_{\mathrm{A}}^{t}\right)}{\left\|g\left(\mathcal{S}_{\mathrm{A}}^{t}\right)\right\|_2 + \varepsilon}, \qquad \lambda_t = \lambda_{\max}\left(\frac{t}{T}\right)^{\gamma} \tag{15}$$

where $\hat{g}(\cdot)$ stabilizes the guidance magnitude, λ_t controls the guidance strength across timesteps, ε avoids division by zero, and ρ balances the repulsion strength from different-class samples. The target-guided synthesis proceeds as follows: 1) obtain a noisy state S_A^T from S_A^0 using (2); 2) for $t = T, T-1, \ldots, 1$, compute $\mu(S_A^t)$ and $\sigma^2(S_A^t)$ from the reverse model, compute $g_{\text{same}}(S_A^t)$ and $g_{\text{diff}}(S_A^t)$, and sample S_A^{t-1} from (13); and 3) output the synthesized sample $\tilde{S}_A^0$ as the generated source training sample used for subsequent cross-scene learning.

Compared with adversarial alignment methods and moment-matching approaches, DADM leverages a generative diffusion process that supplies a smooth noise-to-data path between

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scenes. This progressive mechanism enables gradual and continuous alignment of the source and target scenes, effectively reducing the scene gap while preserving key structural and semantic information from the source scene. Intermediate denoising states serve as natural multi-SNR anchors, helping to maintain class-conditional spectral structures that are crucial for hyperspectral data with multimodal and long-tailed distributions. In addition, DADM incorporates a gradient-based similarity computation, where the gradients of the loss with respect to feature representations guide alignment. This acts as a regularization strategy that promotes domain-invariant feature learning, mitigates adversarial instability, avoids under-alignment of higher order variations, and ultimately improves robustness and generalization in cross-scene classification tasks.

B. Cross-Scene Uncertainty-Aware Mutual Attention Module

After the initial high-dimensional alignment stage, we introduce the CUMAM to refine fine-grained alignment across scenes further. This module is meticulously designed to incorporate class-aware information, which plays a pivotal role in maintaining class consistency between the source and target scenes.

In particular, CUMAM first extracts features from both scenes and projects them into a lower dimensional space, resulting in the representations $\mathcal{S}_0 \in \mathbb{V}^{m\times d}$ and $\mathcal{T}_0 \in \mathbb{V}^{n\times d}$, where d denotes the feature embedding dimension. Inspired by the Transformer architecture, we apply learnable linear projections to map the extracted features into query (Q), key (K), and value (V) matrices as follows:

$$\begin{cases} \mathrm{Q}_{\mathcal{S}} = \mathcal{S}_0\ W_{\mathrm{Q}}^{\mathcal{S}}, \quad \mathrm{K}_{\mathcal{S}} = \mathcal{S}_0\ W_{\mathrm{K}}^{\mathcal{S}}, \quad \mathrm{V}_{\mathcal{S}} = \mathcal{S}_0\ W_{\mathrm{V}}^{\mathcal{S}} \\ \mathrm{Q}_{\mathcal{T}} = \mathcal{T}_0\ W_{\mathrm{Q}}^{\mathcal{T}}, \quad \mathrm{K}_{\mathcal{T}} = \mathcal{T}_0\ W_{\mathrm{K}}^{\mathcal{T}}, \quad \mathrm{V}_{\mathcal{T}} = \mathcal{T}_0\ W_{\mathrm{V}}^{\mathcal{T}}. \end{cases} \tag{16}$$

This transformation enables the construction of a cross-scene attention channel, facilitating bidirectional information exchange between the two scenes. We then compute the attention distributions from source to target and from target to source using scaled dot-product attention

$$\begin{cases} \mathrm{A}^{\mathcal{S}\to\mathcal{T}} = \mathrm{softmax}\left(\dfrac{\mathrm{Q}_{\mathcal{S}}\mathrm{K}_{\mathcal{T}}^{\mathrm{T}}}{\sqrt{\mathrm{d}'}}\right) \\ \mathrm{A}^{\mathcal{T}\to\mathcal{S}} = \mathrm{softmax}\left(\dfrac{\mathrm{Q}_{\mathcal{T}}\mathrm{K}_{\mathcal{S}}^{\mathrm{T}}}{\sqrt{\mathrm{d}'}}\right). \end{cases} \tag{17}$$

Unlike conventional cross-attention mechanisms that solely rely on Q–K–V operations, we further introduce an uncertainty-aware enhancement mechanism to better capture local variations and structural inconsistencies between cross-scene samples.

To this end, we first define the sample-level uncertainty for each sample x_i, calculated as

$$\mathcal{V}(x_i) = \mathrm{sigmoid}\left(\frac{1}{|\mathcal{C}_{x_i}|-1}\sum_{\substack{x_j\in\mathcal{C}_{x_i}\\ j\neq i}}\left\|x_i - x_j\right\|^2 - \frac{1}{\left|\mathcal{C}_{\neg x_i}\right|}\sum_{x_k\in\mathcal{C}_{\neg x_i}}\|x_i - x_k\|^2\right) \tag{18}$$

where $\mathcal{C}x_i$ represents the set of samples belonging to the same class as x_i in a given scene, while $\mathcal{C}\neg x_i$ denotes samples from different classes. The first term captures the intraclass compactness by measuring the average squared distance between x_i and its class peers. The second term quantifies interclass separability by computing the average distance to samples from other categories. A lower score indicates a sample is well-clustered within its class and distant from other classes. To normalize this uncertainty within the range (0, 1), we apply a sigmoid function, allowing it to serve as an attention modulation signal.

In cross-scene classification, this uncertainty score serves as a reliable indicator of sample quality. Samples with lower uncertainty are regarded as reliable features and are expected to guide those with higher uncertainty during the alignment process. A reliable sample is defined as one that demonstrates high intraclass consistency, meaning it is closely clustered with other samples of the same class, and simultaneously exhibits large interclass separability by remaining distant from samples belonging to different classes. In contrast, an unreliable sample is characterized by low intraclass similarity or high proximity to samples of other classes, which often results from label ambiguity, domain shifts, or feature noise. These unreliable samples are more likely to introduce class confusion. To facilitate effective mutual support across scenes, we construct two uncertainty-aware adjustment matrices, denoted as $\mathrm{H}^{\mathcal{S}\to\mathcal{T}} \in \mathbb{V}^{m\times n}$ and $\mathrm{H}^{\mathcal{T}\to\mathcal{S}} \in \mathbb{V}^{n\times m}$, which are defined as follows:

$$\mathrm{H}_{i,j}^{\mathcal{S}\to\mathcal{T}} = \begin{cases} 1 + \dfrac{\mathcal{V}(\mathcal{T}_j)}{\mathcal{V}(\mathcal{S}_i)+\mathcal{V}(\mathcal{T}_j)}, & \mathcal{V}(\mathcal{S}_i) \le \mathcal{V}(\mathcal{T}_j) \\ 1 - \dfrac{\mathcal{V}(\mathcal{T}_j)}{\mathcal{V}(\mathcal{S}_i)+\mathcal{V}(\mathcal{T}_j)}, & \mathcal{V}(\mathcal{S}_i) > \mathcal{V}(\mathcal{T}_j) \\ 1, & \mathrm{cls}(\mathcal{S}_i) \neq \mathrm{cls}(\mathcal{T}_j) \end{cases} \tag{19}$$

$$\mathrm{H}_{i,j}^{\mathcal{T}\to\mathcal{S}} = \begin{cases} 1 + \dfrac{\mathcal{V}(\mathcal{S}_j)}{\mathcal{V}(\mathcal{T}_i)+\mathcal{V}(\mathcal{S}_j)}, & \mathcal{V}(\mathcal{T}_i) \le \mathcal{V}(\mathcal{S}_j) \\ 1 - \dfrac{\mathcal{V}(\mathcal{S}_j)}{\mathcal{V}(\mathcal{T}_i)+\mathcal{V}(\mathcal{S}_j)}, & \mathcal{V}(\mathcal{T}_i) > \mathcal{V}(\mathcal{S}_j) \\ 1, & \mathrm{cls}(\mathcal{T}_i) \neq \mathrm{cls}(\mathcal{S}_j). \end{cases} \tag{20}$$

These matrices serve as dynamic scaling factors that modulate the cross-scene attention distributions based on pairwise uncertainty relationships. Rather than treating all query–key interactions equally, the proposed mechanism adaptively adjusts attention weights by evaluating the relative reliability of each source-target sample pair. This design is motivated by the observation that noisy or ambiguous features can degrade the alignment quality if granted equal influence as more reliable features.

By explicitly incorporating sample-level uncertainty into the attention mechanism, the model achieves several advantages. First, it enhances the robustness of cross-scene alignment by suppressing the impact of unreliable samples, which often arise from domain shift, occlusion, or mislabeling. Second, it promotes a collaborative learning paradigm in which highly discriminative (low-uncertainty) samples play a dominant role in guiding the alignment process, effectively "pulling" the

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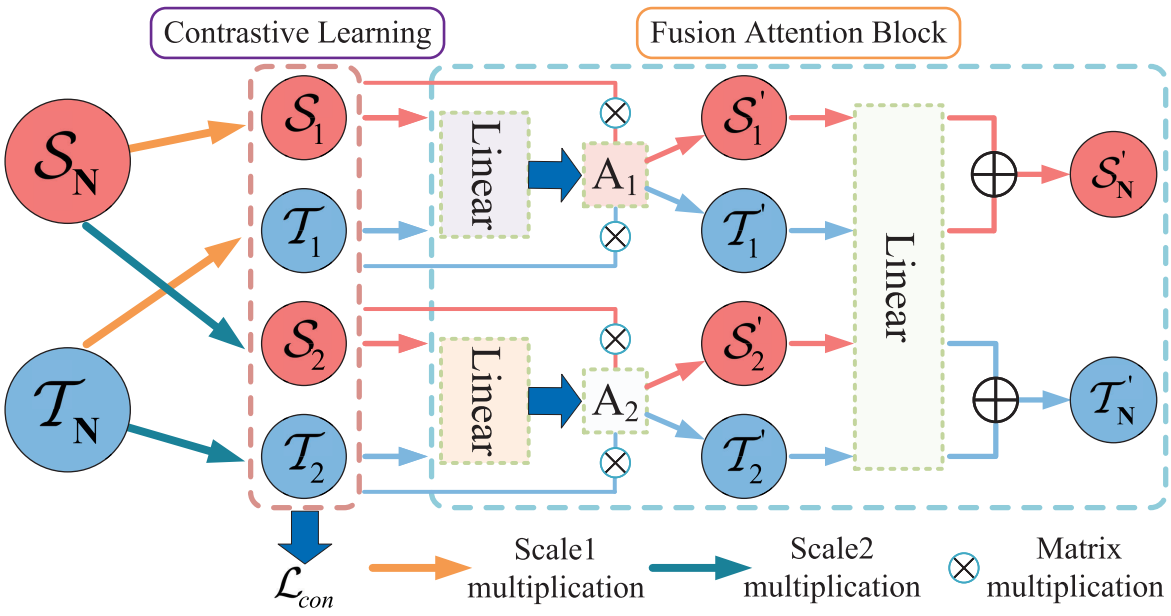


Fig. 2. Architecture of the whole CSMCAB.

uncertain ones toward semantically correct regions in the feature space.

Furthermore, the uncertainty-aware mechanism acts as a soft selection strategy that retains the adaptive nature of attention while incorporating global information about feature reliability. This enables the model to achieve more precise and class-consistent alignment across scenes, thereby enhancing its ability to generalize.

We then apply these factors to reweight the attention distributions, resulting in enhanced mutual attention matrices

$$\widetilde{\mathrm{A}}^{\mathcal{S}\to\mathcal{T}} = \mathrm{A}^{\mathcal{S}\to\mathcal{T}} \odot \mathrm{H}^{\mathcal{S}\to\mathcal{T}}, \widetilde{\mathrm{A}}^{\mathcal{T}\to\mathcal{S}} = \mathrm{A}^{\mathcal{T}\to\mathcal{S}} \odot \mathrm{H}^{\mathcal{T}\to\mathcal{S}}. \quad (21)$$

Finally, we compute the cross-fused outputs for both scenes using the reweighted attention and the corresponding value matrices, followed by a residual connection

$$\mathcal{S}_{\mathrm{N}} = \mathcal{S}_0 + \left(\widetilde{\mathrm{A}}^{\mathcal{S}\to\mathcal{T}} \cdot \mathrm{V}_{\mathcal{T}}\right) \mathrm{W}_1, \mathcal{T}_{\mathrm{N}} = \mathcal{T}_0 + \left(\widetilde{\mathrm{A}}^{\mathcal{T}\to\mathcal{S}} \cdot \mathrm{V}_{\mathcal{S}}\right) \mathrm{W}_2 \quad (22)$$

where W_1 and W_2 are linear transformation matrices that align the dimensionality of the attention outputs with the original features. This residual structure facilitates the integration of both original and cross-aligned features, enabling mutual guidance across scenes and leading to more robust domain generalization. Unlike conventional multiscale or channel/spatial attentions that treat all features uniformly and remain largely scene-agnostic, which often amplify unreliable local information under cross-scene shifts, CUMAM injects uncertainty gating into mutual attention across scenes. This mechanism ensures that alignment is promoted only where correspondences are trustworthy while down-weighting scene-specific artifacts, thereby achieving fine-grained and semantically consistent alignment under local variations.

C. Cross-Scene Fusion Attention Module

To further reduce the feature differences between the two scenes, we integrate multiscale contrastive learning and a cross-scene attention mechanism to construct CSFAM. In particular, CSFAM is implemented by stacking the cross-scene multiscale contrastive attention block (CSMCAB) followed by several linear projection layers for dimensionality reduction and classification. The architecture of the key block CSMCAB is shown in Fig. 2. First, we apply two different dimension reduction standards, extracting features at various scales to better capture global and local relationships. This multiscale approach ensures that the features at different scales capture varying levels of spatial and spectral information, improving the alignment of both fine-grained details (local features) and broader patterns (global features). Subsequently, we introduce the concept of contrastive learning in cross-scene HSI classification. In this context, a positive pair consists of samples that either share the same metric standard or belong to the same class, which are expected to exhibit high similarity. Conversely, a negative pair consists of samples with different metric standards or from different classes, which are expected to demonstrate significant dissimilarity. To formalize this process, we define a contrastive loss function that encourages the model to increase the similarity of positive pairs and the dissimilarity of negative pairs, ensuring a robust feature representation. This loss function is grounded in the theoretical framework of contrastive learning, which aims to reduce intraclass distance while increasing interclass distance, thereby improving the model's ability to distinguish between different classes. It encourages greater similarity within positive pairs and greater dissimilarity within negative pairs. Before constructing this loss function, we first introduce the formula for positive similarity as follows:

$$\mathrm{Sim}_+ = \sum_{i=1}^{m}\sum_{j=1}^{n} k_{ij}^{\mathcal{S}_1,\mathcal{T}_1}\left[\frac{\mathcal{S}_1[i]\cdot\mathcal{T}_1[j]}{\|\mathcal{S}_1[i]\|\,\|\mathcal{T}_1[j]\|} - \frac{\|\mathcal{S}_1[i]-\mathcal{T}_1[j]\|_2}{\max(\mathcal{S}_1,\mathcal{T}_1)}\right] + \sum_{i=1}^{m}\sum_{j=1}^{n} k_{ij}^{\mathcal{S}_2,\mathcal{T}_2}\left[\frac{\mathcal{S}_2[i]\cdot\mathcal{T}_2[j]}{\|\mathcal{S}_2[i]\|\,\|\mathcal{T}_2[j]\|} - \frac{\|\mathcal{S}_2[i]-\mathcal{T}_2[j]\|_2}{\max(\mathcal{S}_2,\mathcal{T}_2)}\right] \quad (23)$$

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where $\mathcal{S}_1 \in \mathbb{V}^{m\times d_1}$ and $\mathcal{T}_1 \in \mathbb{V}^{n\times d_1}$ represent the results of multiplying $\mathcal{S}_\mathbf{N}$ and $\mathcal{T}_\mathbf{N}$ by the first scale, with dimensions $d \times d_1$. Similarly, $\mathcal{S}_2 \in \mathbb{V}^{m\times d_2}$ and $\mathcal{T}_2 \in \mathbb{V}^{n\times d_2}$ represent the results of multiplying by the second scale, with dimensions $d \times d_2$. The expression $\max(s_1, s_2) = \max(|\min(s_1) - \max(s_2)|, |\max(s_1) - \min(s_2)|)$ is used to calculate the value that normalizes the Euclidean distance between two samples. The parameter $k_{ij}^{s_1,s_2}$ is defined in the next formula

$$k_{ij}^{s_1,s_2} = \begin{cases} 0, & \text{cls}\left(s_1\left[i\right]\right) \neq \text{cls}\left(s_2\left[j\right]\right) \\ 1, & \text{cls}\left(s_1\left[i\right]\right) = \text{cls}\left(s_2\left[j\right]\right). \end{cases} \tag{24}$$

Then, we introduce the formula of negative similarity as follows:

$$\begin{aligned} \text{Sim}_- = &\sum_{i=1}^{m}\sum_{j=1}^{n} \bar{k}_{ij}^{\mathcal{S}_1,\mathcal{T}_1}\left[\frac{\mathcal{S}_1\left[i\right]\cdot\mathcal{T}_1\left[j\right]}{\left\|\mathcal{S}_1\left[i\right]\right\|\left\|\mathcal{T}_1\left[j\right]\right\|} - \frac{\left\|\mathcal{S}_1\left[i\right]-\mathcal{T}_1\left[j\right]\right\|_2}{\max\left(\mathcal{S}_1,\mathcal{T}_1\right)}\right] \\ &+ \sum_{i=1}^{m}\sum_{j=1}^{n} \bar{k}_{ij}^{\mathcal{S}_2,\mathcal{T}_2}\left[\frac{\mathcal{S}_2\left[i\right]\cdot\mathcal{T}_2\left[j\right]}{\left\|\mathcal{S}_2\left[i\right]\right\|\left\|\mathcal{T}_2\left[j\right]\right\|} - \frac{\left\|\mathcal{S}_2\left[i\right]-\mathcal{T}_2\left[j\right]\right\|_2}{\max\left(\mathcal{S}_2,\mathcal{T}_2\right)}\right] \end{aligned} \tag{25}$$

where the value $\bar{k}_{ij}$ is defined as $1 - k_{ij}$. The negative similarity function ensures that features from different classes are pushed further apart, facilitating a more discriminative feature space for the classification task. Our goal is to increase the similarity of the positive pairs while making the similarity of the negative pairs more distinct. To achieve this, the contrastive loss $\mathcal{L}_{\text{CON}}$ is defined as follows:

$$\mathcal{L}_{\text{CON}} = -\log \text{P}\left(\frac{\exp\left(\text{Sim}_+/\tau\right)}{\exp\left(\text{Sim}_+/\tau\right) + \exp\left(\text{Sim}_-/\tau\right)}\right). \tag{26}$$

Through contrastive learning, the features from different scenes but with the same scale are initially aligned. This alignment ensures that features from the same class across different scenes maintain high similarity, despite the domain shift. Next, we apply two linear layers with different dimensions, $d_1 \times d_0$ and $d_2 \times d_0$, to obtain the respective attention weights, $\mathbf{A}_1 \in \mathbb{V}^{d_1\times d_0}$ and $\mathbf{A}_2 \in \mathbb{V}^{d_2\times d_0}$. These attention weights are then used in matrix multiplication with the multiscale features as follows:

$$\begin{cases} \mathcal{S}_1' = \mathcal{S}_1\mathbf{A}_1, & \mathcal{T}_1' = \mathcal{T}_1\mathbf{A}_1 \\ \mathcal{S}_2' = \mathcal{S}_2\mathbf{A}_2, & \mathcal{T}_2' = \mathcal{T}_2\mathbf{A}_2. \end{cases} \tag{27}$$

Next, we apply a final linear layer with dimensions $d_0 \times d$ to map the different features back to their original dimensions. We then combine the two source features and the two target features to obtain the final inputs, $\mathcal{S}_\mathbf{N}' \in \mathbb{V}^{m\times d}$ and $\mathcal{T}_\mathbf{N}' \in \mathbb{V}^{n\times d}$. This process helps unify features from different scenes, alleviating domain shift and ensuring feature alignment across scenes. To further reduce the impact of domain shift, we introduce a fusion-invariant loss

$$\begin{aligned} \mathcal{L}_{\text{FI}} = &\sum_{i=1}^{n}\sum_{j=1}^{m} k_{ij}^{\mathcal{T}_1',\mathcal{S}_1'}\left|2\mathcal{T}_1'\left[i\right] - \mathcal{S}_1'\left[j\right] - \mathcal{S}_2'\left[j\right]\right| \\ &+ \sum_{i=1}^{n}\sum_{j=1}^{m} k_{ij}^{\mathcal{T}_2',\mathcal{S}_1'}\left|2\mathcal{T}_2'\left[i\right] - \mathcal{S}_1'\left[j\right] - \mathcal{S}_2'\left[j\right]\right| \\ &+ \left\|\mathcal{S}_1' - \mathcal{S}_2'\right\|_2 + \left\|\mathcal{T}_1' - \mathcal{T}_2'\right\|_2. \end{aligned} \tag{28}$$

The loss function $\mathcal{L}_{\text{FI}}$ consists of two essential components: absolute value terms and L2 regularization terms. The absolute value terms align the source features at different scales with the target features, while the L2 regularization ensures that the features within the same scene are consistent. Together, these components ensure robust feature alignment across scenes while maintaining internal consistency. Finally, by incorporating the original input, the complete CSMCAB architecture is established. After constructing the CSMCAB, several linear layers are applied for dimensionality reduction, leading to the final classification. With these linear layers, the CSFAM is fully completed. CSFAM is not a simple ResNet-like concatenation, but rather a specialized fusion achieved through a cross-scene contrastive learning mechanism combined with a carefully designed loss function. By first separating and then recombining domain-invariant and domain-specific information, it enhances both the complementarity and the consistency of cross-scene features.

D. Loss Function of CDUAN

After constructing the whole CDUAN, we can obtain the loss of CDUAN by the combination as follows:

$$\mathcal{L}_{\text{TOTAL}} = \mathcal{L}_\mathcal{S} + \mathcal{L}_\mathcal{T} + \lambda_1\mathcal{L}_{\text{CON}} + \lambda_2\mathcal{L}_{\text{FI}} \tag{29}$$

where $\mathcal{L}_\mathcal{S}$ and $\mathcal{L}_\mathcal{T}$ denote the cross-entropy losses corresponding to the source and target scenes, respectively. These losses are essential for guiding the model in correctly classifying samples from both scenes. The terms λ_1 and λ_2 represent hyperparameters that are crucial for balancing the contributions of the contrastive loss $\mathcal{L}_{\text{CON}}$ and the fusion-invariant loss $\mathcal{L}_{\text{FI}}$ with the classification losses. The contrastive loss plays a pivotal role in ensuring that the model effectively differentiates between different classes by enhancing the distance between dissimilar pairs and reducing the distance between similar ones. On the other hand, the fusion-invariant loss is designed to promote feature consistency across both source and target scenes, thereby improving the model's generalization capability in cross-scene classification. The combination of these losses in the overall loss function enables the CDUAN model to not only achieve high classification accuracy but also maintain robustness and adaptability across diverse scenes.

IV. Experiments

Extensive experiments are conducted to evaluate the proposed CDUAN, and this section provides a comprehensive overview. First, Section IV-A introduces the three real-world HSI datasets used in the experiments, along with the comparison methods employed. Following that, Section IV-B details the specific experimental settings. In Section IV-C, we present an ablation study that examines the individual components and overall effectiveness of the CDUAN. Finally, Section IV-D offers a thorough comparative analysis, highlighting the performance of CDUAN against various baseline methods.

A. Experimental Datasets and Compared Methods

Three real HSI datasets—RPaviaU-DPaviaC, EHangzhou-RPaviaHR, and Houston—are employed to validate the

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TABLE I
DESCRIPTIONS OF THE THREE DATASETS

Dataset	Scene	Size	Source (Sensor)	Source scene	Target scene
RPaviaU-DPaviaC	RPaviaU	$610 \times 340 \times 103$	University of Pavia (ROSIS)	✓	
	DPaviaC	$400 \times 400 \times 72$	Pavia city (DAIS)		✓
EHangzhou-RPaviaHR	EHangzhou	$590 \times 230 \times 198$	Hangzhou city (EO-1)	✓	
	RPaviaHR	$1400 \times 512 \times 102$	Pavia city (ROSIS)		✓
Houston	Houston2013	$349 \times 1905 \times 144$	Houston (AVIRIS)	✓	
	Houston2018	$601 \times 2384 \times 48$	Houston (LiDAR)		✓

TABLE II
CLASSES AND NUMBER OF SAMPLES IN THE RPAVIAU-DPAVIAC DATASET, THE EHANGZHOU-RPAVIAHR DATASET, AND THE HOUSTON DATASET

EHangzhou-RPaviaHR Dataset				RPaviaU-DPaviaC Dataset			Houston Dataset		
Class		Number		Class	Number		Class	Number	
No.	Name	EHangzhou	RPaviaHR	Name	RPaviaU	DPaviaC	Name	Houston 2013	Houston 2018
1	Water	18403	22525	Trees	3064	2424	Grass healthy	1251	9799
2	Ground/Building	77450	145416	Asphalt	6631	1704	Grass stressed	1254	32502
3	Vegetation	40207	22961	Bitumen	1330	685	Trees	1244	18636
4				Shadow	947	241	Water	325	266
5				Brick	3682	2237	Residential buildings	1268	39762
6				Meadow	18649	1251	Non-residential buildings	1244	223684
7				Soil	5029	1475	Road	1250	45810
Total		136060	190902	Total	39332	10017	Total	7836	370459

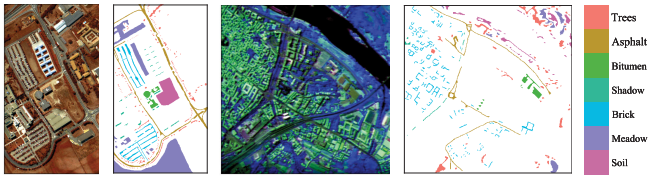


Fig. 3. RPaviaU-DPaviaC dataset. (a) False color (RPaviaU). (b) GroundTruth (RPaviaU). (c) False color (DPaviaC). (d) GroundTruth (DPaviaC). (e) Shared seven land covers.

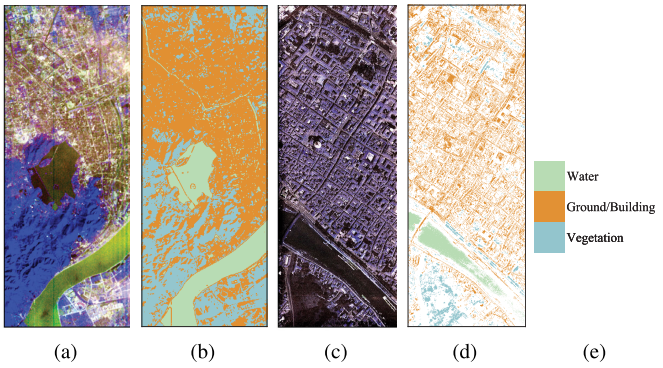


Fig. 4. EHangzhou-RPaviaHR dataset. (a) False color (EHangzhou). (b) GroundTruth (EHangzhou). (c) False color (RPaviaHR). (d) GroundTruth (RPaviaHR). (e) Shared three land covers.

effectiveness of the proposed CDUAN. Table I provides a detailed description of each scene within these datasets. The shared land cover classes and the corresponding number of samples in each dataset are comprehensively listed in Table II. In addition, visual representations of the false color images and their corresponding ground truth maps are thoughtfully presented in Figs. 3–5.

To assess the performance of our proposed CDUAN, we conduct a comparative analysis with several state-of-the-art (SOTA) methods for cross-scene HSI classification. The methods selected for comparison are as follows.

1) *Deep Dynamic Adaptation Network Based on Joint Correlation Alignment (DDAN-JCA):* DDAN-JCA is a deep dynamic adaptation network designed to address the challenges of cross-scene HSI classification [44]. This network employs a joint correlation strategy to capture global correlations between domains and dynamically adjust network parameters to adapt to different scenes, ensuring robust performance across various environments.
2) *Masked Diffusion Model With High-Efficient Transformer (HyperDiff):* The HyperDiff integrates a masked diffusion model with a high-efficiency Transformer to enhance cross-scene HSI classification [45]. It refines noisy features into more discriminative representations while the masked mechanism captures essential spectral–spatial patterns. The efficient Transformer enables global feature learning with reduced computational complexity, ensuring scalability and robustness against domain shifts.
3) *Cross-Domain Discriminative Vision Transformer (CD-DViT):* The CD-DViT addresses heterogeneous cross-scene HSI classification by leveraging self-attention to extract domain-invariant features [46]. An adaptive weighting mechanism dynamically adjusts feature contributions based on domain relevance, reducing domain discrepancy and improving classification accuracy across hyperspectral datasets.
4) *Generative Adversarial Network in Latent Space (GLS):* The GLS is designed to bridge the gap between source and target scenes by constructing a generator that aligns

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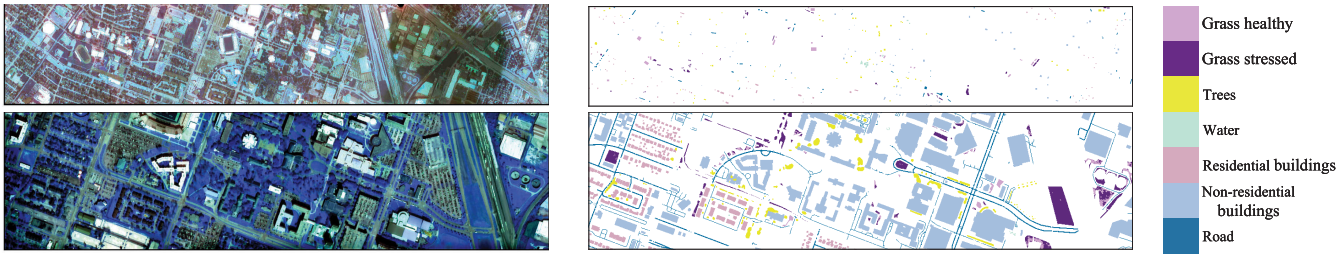


Fig. 5. Houston dataset. (a) False color (the upper represents Houston 2013 and the lower represents Houston 2018). (b) GroundTruth. (c) Shared seven land covers.

latent representations of source scene data with those of target scenes [47]. This approach helps reduce domain discrepancies and improve classification accuracy in cross-scene HSI tasks.

5) *Dual-Stream Discriminative Attention Network (DSDAN):* The DSDAN addresses the challenge of small sample sizes in cross-scene HSI classification [43]. It integrates a dual-stream feature extraction network with a cross-scene attention block and employs a specific loss function to fuse these components effectively. This design enhances the network's ability to discriminate features even in scenarios with limited training data.
6) *Adaptive Graph Modeling With Self-Training (AGM-ST):* The AGM-ST method constructs a graph structure to facilitate information transfer between different scenes [48]. It incorporates a self-training mechanism to generate high-confidence pseudo-labels, which are used to expand the training sample set. This approach enhances the model's adaptability and improves classification accuracy in cross-scene scenarios by effectively utilizing both labeled and pseudo-labeled data.

B. Experimental Settings

In cross-scene classification algorithms, it is essential to tackle real-world scenarios where target training samples are scarce, while source training samples are plentiful. To mimic these conditions, we randomly select 5 samples per class from the target scene and 400 samples per class from the source scene. Our evaluation process involves reporting key metrics, including class-specific accuracy, overall accuracy (OA), average accuracy (AA), and the kappa coefficient (κ). To enhance the robustness of our findings and reduce the impact of random variation, we conduct the experiments across five distinct sample sets, calculating the mean and standard deviation for (OA, AA, and κ) metrics to ensure the reliability and credibility of the results.

The CDUAN involves several hyperparameters that must be fine-tuned to achieve optimal classification performance across different scenes. In particular, the diffusion length T determines the number of reverse denoising steps in the target-guided synthesis process. If T is too small, the reverse process may be under-refined, leading to insufficient denoising and unstable target guidance; if T is excessively large, the accumulated stochasticity and approximation errors may introduce redundant perturbations and increase computational overhead. Moreover, the guidance weight γ controls the strength of the target-guided gradient term in the reverse diffusion, directly affecting how strongly the synthesized samples and learned representations are encouraged to match target-consistent semantics. A small γ may result in weak cross-scene constraints and under-alignment, whereas an overly large γ may over-emphasize guidance and distort class-conditional structures. Besides the diffusion-related parameters, the contrastive learning balance parameter λ_1 adjusts the contribution of the multiscale contrastive objective, shaping the trade-off between intraclass compactness and interclass separation, while the fusion-invariant loss balance parameter λ_2 modulates the strength of fusion-level invariance to suppress scene-dependent feature drift. The sensitivity study and the evaluated settings are summarized in Table III. In particular, we systematically vary the diffusion steps $T \in \{50, 100, 200, 400, 600\}$ and the guidance weight $\gamma \in \{0.2, 0.4, 0.6, 0.8, 1.0\}$. We also explore $\lambda_1, \lambda_2 \in \{0.3, 0.4, 0.5, 0.6, 0.7\}$, reporting OA/AA/$\kappa$ on all three datasets. Based on the empirical results, we adopt the best-performing hyperparameter configuration for each dataset in subsequent experiments: for RPaviaU-DPaviaC, we use ($T = 200, \gamma = 0.4, \lambda_1 = 0.6, \lambda_2 = 0.5$); for EHangzhou-RPaviaHR, we use ($T = 200, \gamma = 0.4, \lambda_1 = 0.5, \lambda_2 = 0.5$); and for Houston, we use ($T = 200, \gamma = 0.4, \lambda_1 = 0.6, \lambda_2 = 0.5$). These dataset-specific settings consistently provide a favorable tradeoff between effectiveness and stability, and are therefore used as the default hyperparameters for all subsequent experiments.

C. Ablation Study on CDUAN

The CDUAN we propose comprises three key components: DADM, CUMAM, and CSMCAB. To demonstrate the contribution of each element, we conducted an ablation study, the results of which are summarized in Table IV. The findings clearly show that each component positively impacts classification accuracy. In particular, the DADM is fundamental, as it generates high-quality source-scene training samples. Its removal leads to a noticeable drop in performance, indicating the importance of clean and informative data for effective DA. The CUMAM introduces a sample-level uncertainty that redefines the reliability of invariant features, assigning closer attention to the more reliable ones to guide the less reliable

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TABLE III
CLASSIFICATION ACCURACIES BETWEEN DIFFERENT PARAMETERS

	RPaviaU-DPaviaC			EHangzhou-RPaviaHR			Houston		
T	OA	AA	κ	OA	AA	κ	OA	AA	κ
50	0.9203	0.9298	0.9034	0.9799	0.9906	0.9502	0.8348	0.8736	0.7418
100	0.9613	0.9605	0.9529	0.9853	0.9935	0.9634	0.8557	0.8862	0.7710
200	**0.9687**	**0.9686**	**0.9619**	**0.9997**	**0.9993**	**0.9992**	**0.8609**	**0.8987**	**0.7806**
400	0.9555	0.9347	0.9457	0.9758	0.9886	0.9406	0.8416	0.8856	0.7543
600	0.9438	0.9416	0.9316	0.9632	0.9710	0.9098	0.8110	0.8778	0.7126
λ_1	OA	AA	κ	OA	AA	κ	OA	AA	κ
0.3	0.9431	0.9400	0.9306	0.9617	0.9830	0.9178	0.8421	0.8984	0.7553
0.4	0.9363	0.9261	0.9223	0.9888	0.9949	0.9718	0.8254	0.8845	0.7305
0.5	0.9378	0.9395	0.9245	**0.9997**	**0.9993**	**0.9992**	0.8525	0.8932	0.7686
0.6	**0.9687**	**0.9686**	**0.9619**	0.9989	0.9991	0.9971	**0.8609**	**0.8987**	**0.7806**
0.7	0.9556	0.9603	0.9459	0.9856	0.9936	0.9641	0.8277	0.8776	0.7326
λ_2	OA	AA	κ	OA	AA	κ	OA	AA	κ
0.3	0.9234	0.9275	0.9070	0.9893	0.9935	0.9729	0.8413	0.8981	0.7549
0.4	0.9422	0.9417	0.9297	0.9942	0.9933	0.9853	0.8243	0.8834	0.7291
0.5	**0.9687**	**0.9686**	**0.9619**	**0.9997**	**0.9993**	**0.9992**	**0.8609**	**0.8987**	**0.7806**
0.6	0.9408	0.9456	0.9280	0.9815	0.9893	0.9539	0.8266	0.8841	0.7329
0.7	0.9144	0.9262	0.8966	0.9772	0.9894	0.9439	0.8385	0.8884	0.7491
γ	OA	AA	κ	OA	AA	κ	OA	AA	κ
0.2	0.9474	0.9532	0.9359	0.9722	0.9877	0.9321	0.8259	0.8825	0.7312
0.4	0.9486	0.9529	0.9375	0.9972	0.9925	0.9929	0.8534	0.8907	0.7764
0.6	**0.9687**	**0.9686**	**0.9619**	**0.9997**	**0.9993**	**0.9992**	**0.8609**	**0.8987**	**0.7806**
0.8	0.9450	0.9483	0.9331	0.9925	0.9910	0.9810	0.8228	0.8685	0.7274
1.0	0.9378	0.9407	0.9244	0.9765	0.9875	0.9419	0.8243	0.8842	0.7304

TABLE IV
ABLATION STUDY ON THREE DATASETS

Method			RPaviaU-DPaviaC			EHangzhou-RPaviaHR			Houston		
DADM	CUMAM	CSMCAB	OA	AA	κ	OA	AA	κ	OA	AA	κ
			0.8656	0.8813	0.8383	0.9359	0.9719	0.8519	0.7481	0.8249	0.6133
✓			0.9215	0.9354	0.9050	0.9790	0.9903	0.9479	0.8242	0.8716	0.7259
	✓		0.9197	0.9155	0.9023	0.9795	0.9908	0.9493	0.8072	0.8638	0.7044
		✓	0.9213	0.9208	0.9045	0.9711	0.9837	0.9292	0.8100	0.8709	0.7068
✓	✓		0.9455	0.9387	0.9337	0.9818	0.9919	0.9549	0.8521	0.8733	0.7629
✓		✓	0.9457	0.9502	0.9352	0.9912	0.9960	0.9779	0.8348	0.8622	0.7407
	✓	✓	0.9411	0.9459	0.9283	0.9869	0.9930	0.9671	0.8331	0.8793	0.7391
✓	✓	✓	**0.9687**	**0.9686**	**0.9619**	**0.9997**	**0.9993**	**0.9992**	**0.8609**	**0.8987**	**0.7806**

representations. Consequently, the alignment between different scenes is effectively strengthened, while removing CUMAM leads to a noticeable degradation in feature alignment performance. CSMCAB promotes feature alignment between the source and target scenes through contrastive learning and cross-scene attention, helping the model distinguish challenging classes, such as those with subtle variations or those underrepresented in the training data. Removing CSMCAB may affect the model's generalization ability, particularly in terms of classification accuracy for these classes. Beyond individual modules, we also examine their combinations. The pairing of DADM and CUMAM yields noticeable performance gains over using either alone, showing that high-quality data and a mutual-attention mechanism jointly enhance adaptation. The combination of DADM and CSMCAB also improves generalization by coupling clean data with strong alignment capabilities. Notably, combining CUMAM and CSMCAB without DADM results in limited improvements, highlighting the foundational role of DADM. Each of these components synergistically enhances the overall effectiveness of the CDUAN model, as demonstrated by the improved results in our ablation study.

D. Comparisons and Analyses

In this section, we provide a detailed analysis of three heterogeneous, cross-scene HSI datasets that involve different sensors and pronounced class-count imbalance, jointly giving rise to fine-grained spectral differences and local boundary variations. We begin with the RPaviaU-DPaviaC dataset. As shown in Table V, the CDUAN achieves the best performance across several classification metrics. Moreover, to help readers clearly identify significant differences, we highlight the incorrectly classified categories in black in Fig. 6. The key findings from this dataset are summarized as follows.

1) In Fig. 6, the highlighted regions indicate that misclassifications of baseline methods mainly happen in narrow transition zones. A representative confusing pair is Asphalt (second class) and Brick (fifth class). By checking typical misclassified samples in the highlighted areas, we find that these errors usually appear around material boundaries where two categories are spatially adjacent, and the local neighborhood contains mixed pixels. In such regions, the spectral responses become very close across several bands, and the local context

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TABLE V

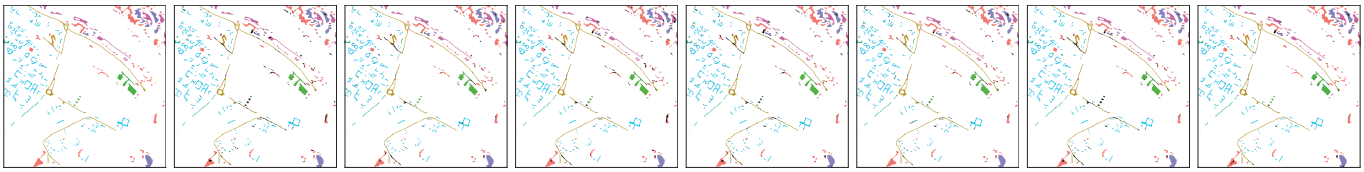
CLASS SPECIFIC AND OA, AA, AND κ OF DIFFERENT METHODS FOR RPAVIAU-DPAVIAC DATASET

Class	DDAN-JCA [44]	HyperDiff [45]	CD-DViT [46]	GLS [47]	DSDAN [43]	AGM-ST [48]	CDUAN
1	0.9516	0.9552	0.8821	0.9487	0.9582	**0.9686**	0.9497
2	0.8658	0.8881	0.9053	0.8328	0.9370	0.9070	**0.9558**
3	0.8897	**0.9493**	0.9373	0.9250	0.9279	0.8559	0.9316
4	0.9915	0.9646	0.9646	0.9703	0.9703	0.9873	**0.9932**
5	0.8526	0.9532	0.9671	0.9297	0.9073	0.9458	**0.9901**
6	0.9791	0.9806	0.9935	0.9719	**0.9880**	0.9711	0.9728
7	0.9061	0.9452	0.9219	0.9313	0.9551	**0.9694**	0.9626
OA	0.9083 ± 0.0035	0.9448 ± 0.0080	0.9305 ± 0.0109	0.9247 ± 0.0039	0.9447 ± 0.0522	0.9462 ± 0.0064	**0.9643 ± 0.0034**
AA	0.9195 ± 0.0032	0.9480 ± 0.0272	0.9388 ± 0.0236	0.9342 ± 0.0108	0.9491 ± 0.0579	0.9436 ± 0.0078	**0.9651 ± 0.0087**
κ	0.8891 ± 0.0028	0.9329 ± 0.0269	0.9157 ± 0.0228	0.9087 ± 0.0183	0.9328 ± 0.0707	0.9345 ± 0.0099	**0.9566 ± 0.0291**

TABLE VI

CLASS SPECIFIC AND OA, AA, AND κ OF DIFFERENT METHODS FOR EHANGZHOU-RPAVIAHR DATASET

Class	DDAN-JCA [44]	HyperDiff [45]	CD-DViT [46]	GLS [47]	DSDAN [43]	AGM-ST [48]	CDUAN
1	**0.9998**	0.9991	0.9994	0.9997	0.9995	0.9992	0.9989
2	0.9031	0.9846	0.9851	0.9280	0.9793	0.9801	**0.9916**
3	0.9987	0.9872	0.9982	0.9942	0.9886	0.9986	**0.9997**
OA	0.9260 ± 0.0047	0.9866 ± 0.0297	0.9884 ± 0.0092	0.9361 ± 0.0256	0.9828 ± 0.0196	0.9845 ± 0.0033	**0.9932 ± 0.0024**
AA	0.9672 ± 0.0092	0.9903 ± 0.0234	0.9942 ± 0.0249	0.9607 ± 0.0102	0.9891 ± 0.0148	0.9926 ± 0.0098	**0.9962 ± 0.0071**
κ	0.8313 ± 0.0128	0.9663 ± 0.0150	0.9708 ± 0.0286	0.8501 ± 0.0136	0.9571 ± 0.0276	0.9616 ± 0.0164	**0.9832 ± 0.0112**



(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 6. Classification maps of RPaviaU-DPaviaC dataset. (a) Ground truth. (b) DDAN-JCA (0.9137). (c) HyperDiff (0.9455). (d) CD-DViT (0.9388). (e) GLS (0.9297). (f) DSDAN (0.9557). (g) AGM-ST (0.9491). (h) CDUAN (0.9687).

is highly heterogeneous, so methods relying on coarse global alignment tend to blur the decision margin and cause boundary leakage. In comparison, CDUAN produces cleaner boundaries and fewer fragmented artifacts in these regions, which is consistent with the diffusion-enhanced adaptation and uncertainty-aware alignment that emphasize discriminative cues around ambiguous local structures.

2) CDUAN consistently achieves the highest class-specific accuracy in three key classes among the SOTA algorithms. In addition, it demonstrates the highest OA, AA, and κ across all SOTAs, indicating that CDUAN effectively balances the learning process and precision, even with limited training samples. In addition, CDUAN exhibits far fewer salt-and-pepper speckles in the classification maps, suggesting better generalization to unseen scenes and greater robustness to scene shift or noise-induced feature drift.

After analyzing the RPaviaU-DPaviaC dataset, we proceed to the EHangzhou-RPaviaHR dataset. Similar to the former, we present the classification results in Table VI, with corresponding classification maps shown in Fig. 7. The key findings for this dataset are summarized below.

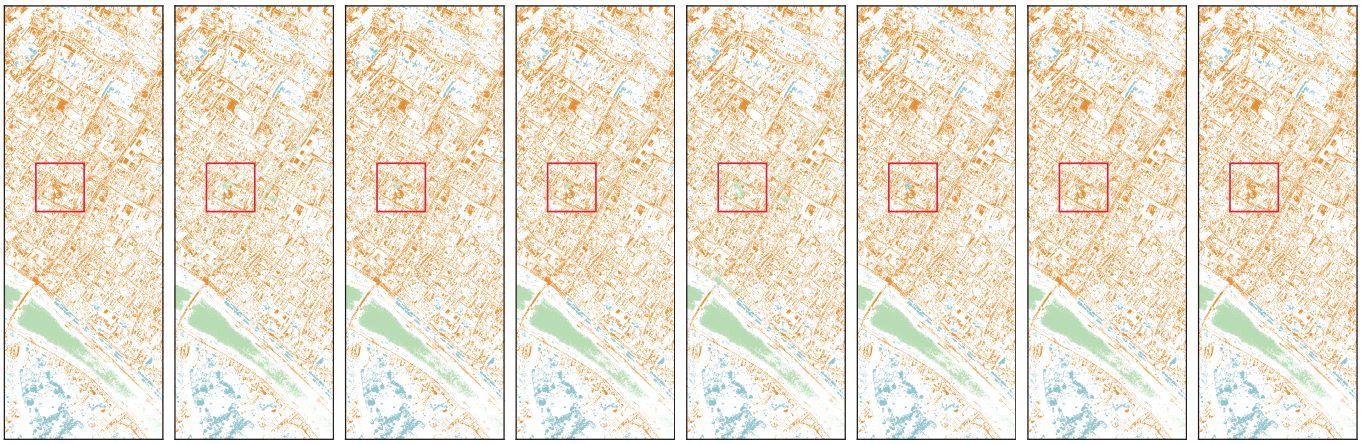
1) Unlike the RPaviaU-DPaviaC dataset, the EHangzhou-RPaviaHR dataset shows that typical misclassified samples are mainly observed at intersections of multiple land-cover types. In these regions, a patch may include several materials while the annotation provides a single label, which makes the supervision intrinsically ambiguous. By contrast, CDUAN separates the intersecting fine-grained classes more consistently and yields a cleaner local structure, indicating that the cross-scene diffusion refinement and uncertainty-guided attention help stabilize predictions in locally heterogeneous and illumination-affected regions.
2) CDUAN achieves the highest accuracy on two of the three class-specific metrics and also ranks best on three overall metrics. In addition, it remains stable across different scene types, indicating strong reliability in diverse and challenging environments. This advantage is further reflected in the red bounding box of the classification map, where other methods exhibit noticeable local confusions, while CDUAN separates the fine-grained classes more cleanly, demonstrating robust adaptation to local variations and improved fine-grained discriminative capability.

Next, we turn to the third dataset, Houston. Table VII provides detailed classification results, while Fig. 8 offers visualizations of the classification outcomes. To help with distinguishing the errors, we highlight the incorrectly classified classes in red. The detailed analysis is as follows.

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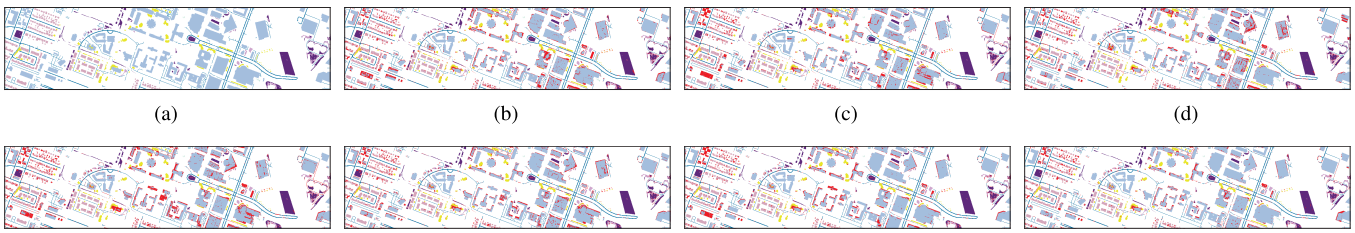
TABLE VII
CLASS SPECIFIC AND OA, AA, AND κ OF DIFFERENT METHODS FOR HOUSTON DATASET

Class	DDAN-JCA [44]	HyperDiff [45]	CD-DViT [46]	GLS [47]	DSDAN [43]	AGM-ST [48]	CDUAN
1	0.9294	0.9190	0.9269	0.8883	0.9288	0.9285	**0.9492**
2	0.9177	**0.9471**	0.9208	0.8727	0.9287	0.9230	0.9397
3	0.8664	0.8979	**0.9178**	0.7649	0.8993	0.9101	0.9172
4	0.9770	0.9947	**0.9948**	0.9847	0.9923	0.9808	0.9866
5	0.7379	0.8134	0.8085	0.7871	0.7752	0.7838	**0.8312**
6	0.8077	0.8230	0.8265	0.7796	0.8392	0.8361	**0.8474**
7	0.8025	0.8379	0.8102	0.8410	**0.8612**	0.8582	0.8488
OA	0.8155 ± 0.0173	0.8411 ± 0.0121	0.8381 ± 0.0054	0.7985 ± 0.0182	0.8485 ± 0.0121	0.8474 ± 0.0203	**0.8542 ± 0.0221**
AA	0.8626 ± 0.0215	0.8904 ± 0.0281	0.8865 ± 0.0084	0.8455 ± 0.0224	0.8893 ± 0.0164	0.8901 ± 0.0264	**0.9015 ± 0.0164**
κ	0.7152 ± 0.0285	0.7532 ± 0.0232	0.7479 ± 0.0106	0.6945 ± 0.0295	0.7632 ± 0.0234	0.7619 ± 0.0152	**0.7721 ± 0.0271**



(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 7. Classification maps of EHangzhou-RPaviaHR dataset. (a) Ground truth. (b) DDAN-JCA (0.9298). (c) HyperDiff (0.9898). (d) CD-DViT (0.9894). (e) GLS (0.9436). (f) DSDAN (0.9903). (g) AGM-ST (0.9936). (h) CDUAN (0.9997).



(a) (b) (c) (d)

(e) (f) (g) (h)

Fig. 8. Classification maps of the Houston dataset. (a) Ground truth. (b) DDAN-JCA (0.8409). (c) HyperDiff (0.8419). (d) CD-DViT (0.8404). (e) GLS (0.8036). (f) DSDAN (0.8597). (g) AGM-ST (0.8519). (h) CDUAN (0.8609).

1) The Houston dataset presents more complex scenes, with many classes being mixed, leading to noisy training patches. This complexity significantly increases the difficulty of accurate classification, resulting in considerable fluctuations depending on the choice of training samples. However, CDUAN delivers the most stable performance because of the cross-scene diffusion model, which effectively minimizes noise and enhances classification accuracy.
2) CDUAN achieves the best class-specific accuracy in four classes and shows significant advantages in distinguishing between residential buildings (class five) and nonresidential buildings (class six). Misclassified samples commonly appear where structural cues are subtle and are weakened by sensor or illumination differences across scenes. Under these conditions, methods without strong cross-scene refinement may rely on insufficient local cues and confuse these closely related classes. CDUAN shows clearer separation on this hard pair and achieves stronger class-wise performance on multiple classes, which is consistent with the diffusion-based adaptation and hierarchical attention design that preserve discriminative details under domain shift.
3) CDUAN maintains robust classification performance under significant domain shifts and scene-specific spectral variability, particularly for residential buildings. This resilience is primarily attributed to the cross-scene diffusion modeling and the integration of adaptive attention modules, which jointly facilitate domain-invariant feature alignment and the suppression of domain-specific

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noise. By selectively enhancing semantically relevant representations and mitigating the influence of redundant or irrelevant information, CDUAN demonstrates superior generalization capability across complex and heterogeneous environments.

In summary, after a comprehensive evaluation of all conducted experiments, the proposed CDUAN consistently outperforms other algorithms, which demonstrates the effectiveness and practical value of CDUAN for cross-scene HSI classification. However, CDUAN still has several limitations, as summarized below.

1) As a target-guided framework, CDUAN requires access to samples from the target scene. Therefore, it does not apply to scenes where no target-scene data are available.
2) CDUAN incurs additional computational overhead due to the diffusion-based component, compared with purely discriminative DA methods.

In future work, we will focus on developing more efficient diffusion strategies and exploring target-free or target-unsupervised sample synthesis to further reduce computational cost and relax the reliance on target-scene supervision.

V. Conclusion

The proposed CDUAN is a DA network that addresses cross-scene HSI classification. DADM enhances the generalization of source training samples by incorporating target scene information, improving cross-scene adaptation. CUMAM aligns features across scenes using a cross-scene mutual attention mechanism, capturing complex spatial–spectral relationships for better scene adaptation. CSFAM ensures feature consistency across scenes by combining contrastive learning with adaptive attention mechanisms, boosting alignment and knowledge transfer. Together, these innovations enable CDUAN to achieve robust, accurate cross-scene learning. Comprehensive experiments on multiple cross-scene datasets confirm the effectiveness of CDUAN, demonstrating its ability to address challenges like limited labeled data in real-world applications.

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